



Faculty of Computing

Year 2 Semester 1 (2025)

IT2120 - Probability and Statistics

Lab Sheet 10

1.) 1. A vending machine owner claims that customers choose the four snack types (A, B, C, D) with equal probability. To test this claim, a researcher records the number of purchases for each snack type during one week and results are given below.

i. State the null and alternative hypotheses for the test.

```
## (i)
## H0: P(A) = P(B) = P(C) = P(D) = 0.25 (choices are equally likely)
## H1: At least one snack type has probability  $\neq$  0.25 (not all equal)

counts <- c(A = 120, B = 95, C = 85, D = 100)
p0 <- rep(1/4, 4)
```

```
> counts <- c(A = 120, B = 95, C = 85, D = 100)
> p0 <- rep(1/4, 4)
> |
```

Environment	History	Connections	Tutorial
  Import Dataset	 133 MiB		
R	Global Environment		
values			
counts	Named num [1:4] 120 95 85 100		
p0	num [1:4] 0.25 0.25 0.25 0.25		

ii. Perform a suitable chi-squared test to test the null hypothesis.

```
# (ii)
chisq_res <- chisq.test(x = counts, p = p0)

cat("Observed counts:\n"); print(counts)
cat("\nExpected counts (under H0):\n"); print(chisq_res$expected)

cat("\nChi-squared statistic:", unname(chisq_res$statistic),
    "\nDegrees of freedom:", unname(chisq_res$parameter),
    "\nP-value:", chisq_res$p.value, "\n")
```

```
> # (ii)
> chisq_res <- chisq.test(x = counts, p = p0)
>
> cat("Observed counts:\n"); print(counts)
Observed counts:
  A   B   C   D
120 95 85 100
> cat("\nExpected counts (under H0):\n"); print(chisq_res$expected)

Expected counts (under H0):
  A   B   C   D
100 100 100 100
>
> cat("\nChi-squared statistic:", unname(chisq_res$statistic),
+     "\nDegrees of freedom:", unname(chisq_res$parameter),
+     "\nP-value:", chisq_res$p.value, "\n")

Chi-squared statistic: 6.5
Degrees of freedom: 3
P-value: 0.0896625
> |
```

Environment	History	Connections	Tutorial
  Import Dataset ▾  134 MiB ▾ 			
R ▾  Global Environment ▾			
Data			
 chisq_res		List of 9	
values			
counts		Named num [1:4] 120 95 85 100	
p0		num [1:4] 0.25 0.25 0.25 0.25	

iii. Give your conclusions based on the results.

```
# (iii)
alpha <- 0.05
if (chisq_res$p.value < alpha) {
  cat("\nDecision: Reject H0 at 5% significance.\n",
      "Conclusion: There IS evidence that snack choices are not equally likely.\n")
} else {
  cat("\nDecision: Fail to reject H0 at 5% significance.\n",
      "Conclusion: There is NOT enough evidence to say snack choices differ from equal likelihood.\n")
}

expected <- sum(counts) * p0
x2_manual <- sum((counts - expected)^2 / expected)
cat("\nManual check of  $\chi^2$ :", x2_manual, "\n")
```

```
> # (iii)
> alpha <- 0.05
> if (chisq_res$p.value < alpha) {
+   cat("\nDecision: Reject H0 at 5% significance.\n",
+       "Conclusion: There IS evidence that snack choices are not equally likely.\n")
+ } else {
+   cat("\nDecision: Fail to reject H0 at 5% significance.\n",
+       "Conclusion: There is NOT enough evidence to say snack choices differ from equal likelihood.\n")
+ }
```

Decision: Fail to reject H0 at 5% significance.
Conclusion: There is NOT enough evidence to say snack choices differ from equal likelihood.

```
>
> expected <- sum(counts) * p0
> x2_manual <- sum((counts - expected)^2 / expected)
> cat("\nManual check of  $\chi^2$ :", x2_manual, "\n")
```

Manual check of χ^2 : 6.5

```
> |
```

Environment	History	Connections	Tutorial	
  Import Dataset ▾	 134 MiB ▾			
R ▾	 Global Environment ▾			
Data				
 chisq_res	List of 9			
values				
alpha	0.05			
counts	Named num [1:4] 120 95 85 100			
expected	num [1:4] 100 100 100 100			
p0	num [1:4] 0.25 0.25 0.25 0.25			
x2_manual	6.5			